

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402413
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Miura; Kiwamu et al.

Display device and electronic apparatus with high density pixels

Abstract

In a display device including a pixel arranged therein, the pixel including a light emitting unit in which an anode electrode is formed in the uppermost layer of a multilayer wiring structure formed by alternately stacking a plurality of insulating layers and a plurality of wiring layers, and a capacitor element electrically connected to the anode electrode of the light emitting unit, or in an electronic device including the display device, the capacitor element including a first electrode formed in a wiring layer below the anode electrode, a second electrode formed opposite to the first electrode, and attached to the anode electrode of the light emitting unit via a conductive rib, and a first insulating layer interposed between the first electrode and the second electrode.

Inventors:	Miura; Kiwamu (Kanagawa, JP), Toyomura; Naobumi (Kanagawa, JP)
Applicant:	Sony Semiconductor Solutions Corporation (Kanagawa, JP)
Family ID:	1000008780799
Assignee:	Sony Semiconductor Solutions Corporation (Kanagawa, JP)
Appl. No.:	18/452910
Filed:	August 21, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240032343 A1	Jan. 25, 2024

Foreign Application Priority Data

JP	2016-174211	Sep. 07, 2016
----	-------------	---------------

Related U.S. Application Data

Publication Classification

Int. Cl.: **H10D86/40** (20250101); **G09F9/30** (20060101); **G09G3/20** (20060101); **G09G3/3233** (20160101); **H05B33/02** (20060101); **H10K50/00** (20230101); **H10K50/11** (20230101); **H10K59/121** (20230101); **H10K59/123** (20230101); **H10K59/80** (20230101)

U.S. Cl.:

CPC **H10D86/481** (20250101); **G09F9/30** (20130101); **G09G3/20** (20130101); **G09G3/3233** (20130101); **H05B33/02** (20130101); **H10K50/00** (20230201); **H10K50/11** (20230201); **H10K59/1213** (20230201); **H10K59/1216** (20230201); **H10K59/123** (20230201); **H10K59/8051** (20230201);

Field of Classification Search

CPC: G09G (3/32); G09G (3/3208); G09G (3/3225); G09G (3/3233); G09G (3/3241); G09G (3/325); G09G (3/3258); H10K (50/00); H10K (50/11); H10K (50/813); H10K (50/814); H10K (59/00); H10K (59/10); H10K (59/12); H10K (59/121); H10K (59/1213); H10K (59/1216); H10K (59/123); H10K (59/8051); H10K (59/80516)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6366025	12/2001	Yamada	N/A	N/A
6630289	12/2002	Kwok et al.	N/A	N/A
6665024	12/2002	Kurashina	N/A	N/A
7696518	12/2009	Koo	N/A	N/A
8570452	12/2012	Moriwaki	N/A	N/A
9142599	12/2014	Koshihara et al.	N/A	N/A
9331133	12/2015	Ota et al.	N/A	N/A
9634079	12/2016	Ota et al.	N/A	N/A
9673267	12/2016	Kim et al.	N/A	N/A
9935163	12/2017	Han et al.	N/A	N/A
10051219	12/2017	Miyake	N/A	H04N 25/76
10128317	12/2017	Wacyk	N/A	N/A
10700212	12/2019	Sawai et al.	N/A	N/A
10777130	12/2019	Ota et al.	N/A	N/A
2008/0074363	12/2007	Yamashita	N/A	N/A
2008/0246403	12/2007	Sagawa et al.	N/A	N/A
2014/0217432	12/2013	Ota	N/A	N/A
2014/0291636	12/2013	Kim et al.	N/A	N/A
2015/0069363	12/2014	Ota	N/A	N/A
2015/0097172	12/2014	Han	N/A	N/A
2016/0064467	12/2015	Ota	257/40	H10K 59/1216

2016/0191825	12/2015	Sato et al.	N/A	N/A
2016/0197130	12/2015	Gil et al.	N/A	N/A
2018/0240834	12/2017	Tanaka et al.	N/A	N/A
2022/0271086	12/2021	Kong	N/A	H10H 29/142

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101286524	12/2007	CN	N/A
101404140	12/2008	CN	N/A
102246128	12/2010	CN	N/A
102447074	12/2011	CN	N/A
102623485	12/2011	CN	N/A
203659396	12/2013	CN	N/A
104517996	12/2014	CN	N/A
105390527	12/2015	CN	N/A
113270464	12/2020	CN	N/A
113314574	12/2020	CN	N/A
2005227562	12/2004	JP	N/A
2008287135	12/2007	JP	N/A
2011-221071	12/2010	JP	N/A
2013044890	12/2012	JP	N/A
2014085385	12/2013	JP	N/A
2014186139	12/2013	JP	N/A
2015191038	12/2014	JP	N/A
2016-053636	12/2015	JP	N/A
2016-053640	12/2015	JP	N/A
2016045304	12/2015	JP	N/A

OTHER PUBLICATIONS

Japanese Office Action issued Jan. 12, 2021 for corresponding Japanese Application No. 2018-538272. cited by applicant

Chinese Office Action dated Nov. 9, 2020 for corresponding Chinese Application No. 2017800530796. cited by applicant

Japanese Office Action issued Mar. 23, 2021 for corresponding Japanese Application No. 2018-538272. cited by applicant

Primary Examiner: Partridge; William B

Assistant Examiner: Ramallo; Gustavo G

Attorney, Agent or Firm: Michael Best & Friedrich LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a Continuation application of U.S. patent application Ser. No. 16/322,008, filed Jan. 30, 2019, which is a 371 National Stage Entry of International Application No.: PCT/JP2017/026890, filed on Jul. 25, 2017, which in turn claims priority from Japanese Priority Patent Application JP 2016-174211 filed on Sep. 7, 2016, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a display device and an electronic device including the display device.

BACKGROUND ART

(2) As one of display devices, there is an organic electroluminescence (hereinafter, referred to as “EL”) display device using an organic EL element using an EL that is an organic material as a light emitting unit. The organic EL display device has a lower power consumption characteristic, since the organic EL element forming the light emitting unit is a self-light emitting element. The organic EL element has a structure in which an organic layer formed by stacking an organic hole transporting layer and an organic light emitting layer is provided between a first electrode and a second electrode. Then, in the organic EL display device, gradation of coloring is obtained by controlling the current value flowing through the organic EL element.

(3) Among organic EL display devices using an organic EL element as a light emitting unit, in an active matrix type organic EL display device, a drive circuit (pixel circuit) including a plurality of transistors and a capacitor element is provided for each pixel, and driving of the organic EL element is performed by this drive circuit. Then, in order to effectively utilize the upper layer than a gate electrode of the transistor to provide a pixel structure for high density pixels, a capacitor element is formed using a wiring layer right above the gate electrode (see, for example, Patent Document 1).

CITATION LIST

Patent Document

(4) Patent Document 1: Japanese Patent Application Laid-Open No. 2016-53640

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(5) Meanwhile, in ultra-compact display devices in which an interval between pixels decreases due to higher definition, or ultra-compact display devices called micro display, since the pixel size becomes small, it is difficult to sufficiently secure a capacitance value of a capacitor element. In particular, in a wiring layer right above a gate electrode of a transistor, since a capacitor element is formed avoiding other gate wiring or the like, it is difficult to efficiently form a capacitor element. As a result, there is a problem that the display quality is deteriorated due to insufficient capacity of the capacitor element.

(6) In view of the above, an object of the present disclosure is to provide a display device capable of suppressing deterioration in display quality due to insufficient capacity of a capacitor element by efficiently forming a capacitor element and sufficiently securing a capacitance value of the capacitor element, and an electronic device including the display device.

Solutions to Problems

(7) A display device of the present disclosure for achieving the object described above includes a pixel arranged therein, the pixel including a light emitting unit in which an anode electrode is formed in the uppermost layer of a multilayer wiring structure formed by alternately stacking a plurality of insulating layers and a plurality of wiring layers, and a capacitor element electrically connected to the anode electrode of the light emitting unit, the capacitor element including a first electrode formed in a wiring layer below the anode electrode, a second electrode formed opposite to the first electrode, and electrically connected to the anode electrode of the light emitting unit via a contact unit, and a first insulating layer interposed between the first electrode and the second electrode. Furthermore, an electronic device of the present disclosure for achieving the object described above is characterized by including the display device having the above-described configuration.

(8) In the display device having the above configuration or the electronic device including the display device, the first electrode is formed in a wiring layer below the anode electrode of the light

emitting unit, and the second electrode is electrically connected to the anode electrode of the light emitting unit via the contact unit, so that a region below the anode electrode of the light emitting unit can be efficiently used to form the capacitor element. Furthermore, the length of the contact unit is adjusted and an interval between the first electrode and the second electrode is arbitrarily set, so that a desired capacitance value can be easily secured as a capacitance value of the capacitor element.

Effects of the Invention

(9) According to the present disclosure, it is possible to suppress deterioration in display quality due to insufficient capacity of a capacitor element by efficiently forming a capacitor element and sufficiently securing a capacitance value of the capacitor element.

(10) Note that, the effect is not necessarily limited to the effect described herein, and any of the effects described in this specification may be used. Furthermore, the effects described in this specification are merely examples, and the present invention is not limited thereto, and may have additional effects.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a system configuration diagram schematically showing a configuration of an active matrix type organic EL display device of the present disclosure.

(2) FIG. 2 is a circuit diagram showing a circuit configuration of a pixel (pixel circuit) of 2Tr1C in the active matrix type organic EL display device of the present disclosure.

(3) FIG. 3 is a cross-sectional view showing a sectional structure of a holding capacitor according to a conventional example.

(4) FIG. 4 is a cross-sectional view showing a sectional structure of a holding capacitor according to a first embodiment.

(5) FIG. 5 is a cross-sectional view showing a sectional structure of a holding capacitor according to a second embodiment.

(6) FIG. 6 is a cross-sectional view showing a sectional structure of a holding capacitor according to a third embodiment.

(7) FIG. 7A and FIG. 7B are a front view and a rear view of a lens interchangeable single lens reflex type digital still camera.

(8) FIG. 8 is an external view of a head mounted display.

MODE FOR CARRYING OUT THE INVENTION

(9) Hereinafter, modes (hereinafter, referred to as “embodiments”) for implementing the technology of the present disclosure will be described in detail with reference to the drawings. The technology of the present disclosure is not limited to the embodiments. In the following description, the same reference numerals will be used for the same elements or elements having the same function, and redundant description will be omitted. Note that the description will be given in the following order. 1. General explanation of display device and electronic device of the present disclosure 2. Active matrix type organic EL display device 2-1. System configuration 2-2. Pixel circuit 2-3. Holding capacitor 2-3-1. Example of forming capacitor element using wiring layer right above gate electrode 2-3-2. First embodiment 2-3-3. Second embodiment (modification of first embodiment) 2-3-4. Third embodiment (modification of second embodiment) 3. Modification 4. Electronic device of the present disclosure 4-1. Specific example 1 (example of digital still camera) 4-2. Specific example 2 (example of head mounted display) 5. Configuration that the present disclosure can have

<General Explanation of Display Device and Electronic Device of the Present Disclosure>

(10) In the display device and the electronic device of the present disclosure, the capacitor element

can include a third electrode formed in a wiring layer below the first electrode, a fourth electrode formed opposite to the third electrode and electrically connected to the first electrode via a contact unit, and a second insulating layer interposed between the third electrode and the fourth electrode. At this time, it is preferable that the third electrode be electrically connected to the anode electrode and the fourth electrode be electrically connected to the second electrode.

(11) In the display device, the driving method of the display device, and the electronic device of the present disclosure having the preferable configuration described above, the pixels can have a driving transistor for driving the light emitting unit and a writing transistor for writing a signal to the gate electrode of the driving transistor. Furthermore, the capacitor element can be connected between the gate electrode of the driving transistor and the anode electrode of light emitting unit, and hold the signal written by the writing transistor.

(12) Moreover, in the display device, a driving method of the display device of the present disclosure having the preferable configuration described above, and the electronic device, a substrate on which the pixels are two-dimensionally arranged can be a semiconductor substrate. Furthermore, the light emitting unit can include an organic electroluminescence element.

(13) <Active Matrix Type Display Device>

(14) The display device of the present disclosure is an active matrix type display device in which a current flowing through an electrooptic element is controlled by an active element provided in the same pixel circuit as the electrooptic element, for example, an insulated gate field effect transistor. As the insulated gate type field effect transistor, typically, a metal oxide semiconductor (MOS) transistor and a thin film transistor (TFT) can be exemplified.

(15) Here, described as an example is an active matrix type organic EL display device using an organic EL element that is a current driven type electrooptic element whose light emission luminance changes according to a current value flowing in the device, as a light emitting unit (light emitting element) of a pixel circuit. Hereinafter, the “pixel circuit” may be simply described as “pixel”.

(16) [System Configuration]

(17) FIG. 1 is a system configuration diagram schematically showing a configuration of the active matrix type organic EL display device of the present disclosure. As shown in FIG. 1, an organic EL display device **10** of the present disclosure includes a pixel array unit **30** in which a plurality of pixels **20** including organic EL elements are two-dimensionally arranged in a matrix (matrix shape), and a peripheral circuit unit arranged in a periphery of the pixel array unit **30**. The peripheral circuit unit includes, for example, a writing scanning unit **40**, a power supply scanning unit **50**, a signal output unit **60**, and the like, and drives each pixel **20** of the pixel array unit **30**.

(18) In this example, the writing scanning unit **40**, the power supply scanning unit **50**, and the signal output unit **60** are mounted on the same substrate as the pixel array unit **30**, in other words, on a display panel **70** as a peripheral circuit of the pixel array unit **30**. However, any or all of the writing scanning unit **40**, the power supply scanning unit **50**, and the signal output unit **60** can be provided outside the display panel **70**. Furthermore, although the writing scanning unit **40** and the power supply scanning unit **50** are arranged on one side of the pixel array unit **30**, the writing scanning unit **40** and the power supply scanning unit **50** can be arranged on both sides of the pixel array unit **30**.

(19) The organic EL display device **10** can be configured to be compatible with monochrome (black and white) display, or can be configured to be compatible with color display. In a case where the organic EL display device **10** is compatible with color display, one pixel (unit pixel/pixel) as a unit for forming a color image includes a plurality of sub pixels. At this time, each of the subpixels corresponds to the pixel **20** in FIG. 1. More specifically, in a display device compatible with color display, one pixel includes, for example, three sub pixels, that is, a sub pixel that emits red (R) light, a sub pixel that emits green (G) light, and a sub pixel that emits blue (B) light.

(20) However, the one pixel is not limited to a combination of sub pixels of three primary colors of

RGB, and it is also possible to form one pixel by further adding sub pixels of one or more colors to the sub pixels of three primary colors. More specifically, for example, in order to improve the luminance, a sub pixel emitting white (W) light can be added to form one pixel, or at least one sub pixel emitting complementary color light for enlarging the color reproduction range can be added to form one pixel.

(21) In the pixel array unit **30**, scanning lines **31** (**31.sub.1** to **31.sub.m**), and power supply lines **32** (**32.sub.1** to **32.sub.m**) are wired for each pixel row along the row direction (pixel array direction of pixel rows) with respect to the array of the pixels **20** of m rows and n columns. Moreover, signal lines **33** (**33.sub.1** to **33.sub.n**) are wired for each pixel column along the column direction (pixel array direction of pixel columns) with respect to the array of the pixels **20** of m rows and n columns.

(22) The scanning lines **31.sub.1** to **31.sub.m** are respectively connected to output ends of the corresponding rows of the writing scanning unit **40**. The power supply lines **32.sub.1** to **32.sub.m** are respectively connected to output ends of the corresponding rows of the power supply scanning unit **50**. The signal lines **33.sub.1** to **33.sub.n** are respectively connected to output ends of the corresponding columns of the signal output unit **60**.

(23) The writing scanning unit **40** includes a shift register circuit or the like. When writing a signal voltage of a video signal to each pixel **20** of the pixel array unit **30**, the writing scanning unit **40** sequentially supplies writing scanning signals WS (**WS.sub.1** to **WS.sub.m**) to the scanning lines **31** (**31.sub.1** to **31.sub.m**), so that each pixel **20** of the pixel array unit **30** is sequentially scanned in units of row, that is, so-called line sequential scanning is performed.

(24) As similar to the writing scanning unit **40**, the power supply scanning unit **50** includes a shift register circuit or the like. In synchronism with the line sequential scanning by the writing scanning unit **40**, the power supply scanning unit **50** supplies a power supply voltage DS (**DS.sub.1** to **DS.sub.m**) capable of switching between the first power supply voltage **V.sub.ccp** and the second power supply voltage **V.sub.ini** lower than the first power supply voltage **V.sub.ccp** to the power supply lines **32** (**32.sub.1** to **32.sub.m**). As described later, light emission/non-light emission (extinction) of the pixel **20** is controlled by switching **V.sub.ccp/V.sub.ini** of the power supply voltage DS.

(25) The signal output unit **60** selectively outputs a signal voltage (hereinafter, sometimes referred to simply as “signal voltage”) **V.sub.sig** of a video signal corresponding to luminance information supplied from a signal supply source (not shown), and a reference voltage **V.sub ofs**. Here, the reference voltage **V.sub ofs** is a voltage as a reference of the signal voltage **V.sub.sig** of the video signal (for example, a voltage corresponding to the black level of the video signal), and is used in the threshold correction processing as described later.

(26) The signal voltage **V.sub.sig**/reference voltage **V.sub ofs** output from the signal output unit **60** is written to each pixel **20** of the pixel array unit **30** via the signal lines **33** (**33.sub.1** to **33.sub.n**) in units of pixel rows selected by scanning by the writing scanning circuit **40**. In other words, the signal output unit **60** adopts a line sequential writing drive mode in which the signal voltage **V.sub.sig** is written in units of rows (lines).

(27) [Pixel Circuit]

(28) FIG. 2 is a circuit diagram showing an example of a specific circuit configuration of the pixel (pixel circuit) **20**. The light emitting unit of the pixel **20** includes the organic EL element **21** being an example of a current driven type electrooptic element in which the light emission luminance changes according to a current value flowing in the device.

(29) As shown in FIG. 2, the pixel **20** includes an organic EL element **21** and a drive circuit for driving the organic EL element **21** by applying a current to the organic EL element **21**. In the organic EL element **21**, a cathode electrode is connected to a common power supply line **34** wired in common for all the pixels **20**.

(30) The driving circuit for driving the organic EL element **21** has a circuit configuration, so-called

2Tr1C, including the driving transistor **22**, the writing transistor **23**, and the holding capacitor **24**, in other words, two transistors (Tr) and one capacitor element (C). Here, an N-channel thin film transistor (TFT) is used as the driving transistor **22** and the writing transistor **23**. However, the combination of the conductivity types of the driving transistor **22** and the writing transistor **23** shown here is a merely example, and the combination is not limited thereto.

(31) Note that, in the organic EL display device **10** of the present disclosure, the pixel (pixel circuit) **20** is formed not on an insulator such as a glass substrate but on a semiconductor such as silicon. In other words, the substrate on which the pixels **20** are two-dimensionally arranged is a semiconductor substrate.

(32) In the driving transistor **22**, one electrode (source/drain electrode) is connected to the power supply lines **32** (**32.sub.1** to **32.sub.m**) and another electrode (source/drain electrode) is connected to the anode electrode of the organic EL element **21**. In the writing transistor **23**, one electrode (source/drain electrode) is connected to the signal lines **33** (**33.sub.1** to **33.sub.n**) and another electrode (source/drain electrode) is connected to the gate electrode of the driving transistor **22**. Furthermore, the gate electrode of the writing transistor **23** is connected to the scanning lines **31** (**31.sub.1** to **31.sub.m**).

(33) In the driving transistor **22** and the writing transistor **23**, one electrode is a metal wiring electrically connected to one source/drain region, and another electrode is a metal wiring electrically connected to another source/drain region. Furthermore, depending on the potential relationship between one electrode and another electrode, one electrode may be the source electrode or the drain electrode in some cases, and another electrode may be the drain electrode or the source electrode in some cases.

(34) One electrode of the holding capacitor **24** is connected to the gate electrode of the driving transistor **22** and another electrode is connected to another electrode of the driving transistor **22** and to the anode electrode of the organic EL element **21**.

(35) In the above configuration, the writing transistor **23** enters a conductive state in response to the writing scanning signal WS that is applied to the gate electrode from the writing scanning unit **40** through the scanning line **31** and in which the high voltage state corresponds to an active state. As a result, the writing transistor **23** samples the signal voltage V.sub.sig or the reference voltage V.sub ofs of the video signal corresponding to the luminance information that is supplied from the signal output unit **60** at different timings through the signal line **33**, and writes the signal voltage V.sub.sig or the reference voltage V.sub ofs into the pixel **20**. The signal voltage V.sub.sig or the reference voltage V.sub ofs written by the writing transistor **23** is held in the holding capacitor **24**.

(36) When the power supply voltage DS of the power supply lines **32** (**32.sub.1** to **32.sub.m**) is at the first power supply voltage V.sub.ccp, the driving transistor **22** operates in a saturation region with one electrode serving as a drain electrode and another electrode serving as a source electrode. As a result, the driving transistor **22** receives a current supply from the power supply line **32** and drives the organic EL element **21** to emit light by current driving. More specifically, by operating in the saturation region, the driving transistor **22** supplies a driving current of a current value corresponding to the voltage value of the signal voltage V.sub.sig held in the holding capacitor **24** to the organic EL element **21**, and drives the organic EL element **21** by current to emit light.

(37) Moreover, when the power supply voltage DS is switched from the first power supply voltage V.sub.ccp to the second power supply voltage V.sub.ini, the driving transistor **22** operates as a switching transistor with one electrode serving as a source electrode and another electrode serving as a drain electrode. As a result, the driving transistor **22** stops the supply of the driving current to the organic EL element **21**, and sets the organic EL element **21** to the non-light emitting state. In other words, the driving transistor **22** also has a function as a transistor for controlling light emission/non-light emission of the organic EL element **21**.

(38) By the switching operation of the driving transistor **22**, a period (non-light emitting period) during which the organic EL element **21** is in the non-light emitting state can be provided to control

the ratio (duty) of the light emitting period and the non-light emitting period of the organic EL element **21**. By this duty control, afterimage blurring caused by light emission of the pixels over one display frame period can be reduced, so that, in particular, image quality of moving images can be made more excellent.

(39) Among the first power supply voltage $V_{\text{sub.ccp}}$ and the second power supply voltage $V_{\text{sub.ini}}$ selectively supplied from the power supply scanning unit **50** through the power supply line **32**, the first power supply voltage $V_{\text{sub.ccp}}$ is a power supply voltage for supplying the driving current for driving the organic EL element **21** to emit light to the driving transistor **22**. Furthermore, the second power supply voltage $V_{\text{sub.ini}}$ is a power supply voltage for applying a reverse bias to the organic EL element **21**. The second power supply voltage $V_{\text{sub.ini}}$ is set to a voltage lower than the reference voltage $V_{\text{sub ofs}}$, for example, when the threshold voltage of the driving transistor **22** is $V_{\text{sub.th}}$, a voltage lower than $V_{\text{sub ofs}} - V_{\text{sub.th}}$, preferably a voltage sufficiently lower than $V_{\text{sub ofs}} - V_{\text{sub.th}}$.

(40) Each of the pixels **20** of the pixel array unit **30** has a function of correcting variation in driving current caused by variation in characteristics of the driving transistor **22**. As the characteristics of the driving transistor **22**, for example, the threshold voltage $V_{\text{sub.th}}$ of the driving transistor **22**, and the mobility μ of the semiconductor thin film included in the channel of the driving transistor **22** (hereinafter simply referred to as “mobility μ of the driving transistor **22**”) can be exemplified.

(41) The correction of the variation in the driving current caused by the variation in the threshold voltage $V_{\text{sub.th}}$ (hereinafter referred to as “threshold correction”) is performed by initializing the gate voltage $V_{\text{sub.g}}$ of the driving transistor **22** to the reference voltage $V_{\text{sub ofs}}$. Specifically, with the initialization voltage (reference voltage $V_{\text{sub ofs}}$) of the gate voltage $V_{\text{sub.g}}$ of the driving transistor **22** taken as a reference, the operation of changing the source voltage $V_{\text{sub.s}}$ of the driving transistor **22** toward the potential obtained by subtracting the threshold voltage $V_{\text{sub.th}}$ of the driving transistor **22** from the initialization voltage is performed. As the operation progresses, the gate-source voltage $V_{\text{sub.gs}}$ of the driving transistor **22** eventually converges to the threshold voltage $V_{\text{sub.th}}$ of the driving transistor **22**. A voltage corresponding to this threshold voltage $V_{\text{sub.th}}$ is held in the holding capacitor **24**. Then, since the voltage corresponding to the threshold voltage $V_{\text{sub.th}}$ is held in the holding capacitor **24**, when the driving transistor **22** is driven by the signal voltage $V_{\text{sub.sig}}$ of the video signal, dependency on the threshold voltage $V_{\text{sub.th}}$ of the drain-source current $I_{\text{sub.ds}}$ flowing in the driving transistor **22** can be suppressed.

(42) On the other hand, the correction of the variation in the driving current caused by the variation in the mobility μ (hereinafter, referred to as “mobility correction”) is performed by flowing the current via the driving transistor **22** to the holding capacitor **24** in a state where the writing transistor **23** is in the conductive state and the signal voltage $V_{\text{sub.sig}}$ of the video signal is being written. In other words, the mobility correction is performed by applying negative feedback to the holding capacitor **24** with a feedback amount (correction amount) corresponding to the current $I_{\text{sub.ds}}$ flowing in the driving transistor **22**. Through the above-described threshold correction, when the video signal is written, the dependence on the threshold voltage $V_{\text{sub.th}}$ of the drain-source current $I_{\text{sub.ds}}$ has already been canceled, and the drain-source current $I_{\text{sub.ds}}$ depends on the mobility μ of the driving transistor **22**. Accordingly, negative feedback is applied to the drain-source voltage $V_{\text{sub.ds}}$ of the driving transistor **22** with a feedback amount corresponding to the current $I_{\text{sub.ds}}$ flowing in the driving transistor **22**, so that the dependence on the mobility μ of the drain-source current $I_{\text{sub.ds}}$ flowing in the driving transistor **22** can be suppressed.

(43) [Holding Capacitor]

(44) In the above-described active matrix type organic EL display device **10**, as the number of pixels increases with higher definition, the size of the pixels **20** decreases. Furthermore, in a microdisplay (ultra-compact display device), as a matter of course, the size of the pixels **20** is smaller than that in a general display for application as a monitor. As described above, when the size of the pixels **20** is small, it is difficult to sufficiently secure the capacitance value of the

holding capacitor **24**.

(45) As described above, the holding capacitor **24** is a capacitor element provided to hold the signal voltage $V_{\text{sub.sig}}$ written by the writing transistor **23**. Accordingly, if the capacitance value of the holding capacitor **24** cannot be secured, the signal voltage $V_{\text{sub.sig}}$ written by the writing transistor **23** cannot be sufficiently held.

(46) Furthermore, the capacitance value of the holding capacitor **24** can also affect bootstrap operation for keeping the light emission luminance of the organic EL element **21** constant even if the I (current) – V (voltage) characteristic of the organic EL element **21** varies with time. The “bootstrap operation” is operation in which the gate voltage $V_{\text{sub.g}}$ of the driving transistor **22** fluctuates in conjunction with fluctuation of the source voltage $V_{\text{sub.s}}$, in other words, operation in which, while the gate-source voltage $V_{\text{sub.gs}}$ of the driving transistor **22** is kept constant, the gate voltage $V_{\text{sub.g}}$ and the source voltage $V_{\text{sub.s}}$ rise.

(47) Here, the capacitance value of the holding capacitor **24** is defined as $C_{\text{sub.s}}$, the parasitic capacitance between the gate and the source of the driving transistor **22** is defined as $C_{\text{sub.gs}}$, the parasitic capacitance between the gate and the drain of the driving transistor **22** is defined as $C_{\text{sub.gd}}$, and the parasitic capacitance between the gate-drain and the source of the writing transistor **23** is defined as $C_{\text{sub.d}}$. Furthermore, the fluctuation component of the source voltage $V_{\text{sub.s}}$ of the driving transistor **22** is defined as $\Delta V_{\text{sub.s}}$, and the fluctuation component of the gate voltage $V_{\text{sub.g}}$ is defined as $\Delta V_{\text{sub.g}}$. Then, the ratio of the fluctuation component $\Delta V_{\text{sub.g}}$ of the gate voltage $V_{\text{sub.g}}$ with respect to the fluctuation component $\Delta V_{\text{sub.s}}$ of the source voltage $V_{\text{sub.s}}$ of the driving transistor, in other words, the bootstrap ratio $G_{\text{sub.bst}}$ is represented by $G_{\text{sub.bst}} = \Delta V_{\text{sub.g}} / \Delta V_{\text{sub.s}}$

$$= \{(C_{\text{sub.s}} + C_{\text{sub.gs}}) / (C_{\text{sub.s}} + C_{\text{sub.gs}} + C_{\text{sub.gd}} + C_{\text{sub.d}})\} \quad (1).$$

(48) In the bootstrap operation, if the bootstrap ratio $G_{\text{sub.bst}}$ is small, the fluctuation component $\Delta V_{\text{sub.g}}$ of the gate voltage $V_{\text{sub.g}}$ is smaller than the fluctuation component $\Delta V_{\text{sub.s}}$ of the source voltage $V_{\text{sub.s}}$, so that the gate-source voltage $V_{\text{sub.gs}}$ is reduced. As a result, the gate-source voltage $V_{\text{sub.gs}}$ becomes smaller than the signal voltage $V_{\text{sub.sig}}$ held in the holding capacitor **24** by writing by the writing transistor **23**. Then, it is impossible to secure a current required as a current to be supplied to the organic EL element **21**, in other words, a current corresponding to the signal voltage $V_{\text{sub.sig}}$ written by sampling by the writing transistor **23**. As a result, the luminance decreases, and accordingly display unevenness occurs, which results in deterioration in display quality.

(49) As apparent from the above equation (1), the bootstrap ratio $G_{\text{sub.bst}}$ is determined by the capacitance value $C_{\text{sub.s}}$ of the holding capacitor **24** and the parasitic capacitances $C_{\text{sub.gs}}$, $C_{\text{sub.gd}}$, $C_{\text{sub.d}}$ attached to the gate electrode of the driving transistor **22**. Accordingly, if the capacitance value $C_{\text{sub.s}}$ of the holding capacitor **24** is large, the bootstrap ratio $G_{\text{sub.bst}}$ becomes large, so that it is necessary to secure a large capacitance value $C_{\text{sub.s}}$ of the holding capacitor **24** from the viewpoint of display quality.

(50) As shown in FIG. 3, the organic EL display device **10** according to the present embodiment has a multilayer wiring structure formed by alternately stacking a plurality of insulating layers, for example, four insulating layers **71** to **74**, and a plurality of wiring layers, for example, four wiring layers **81** to **84**. Then, an anode electrode $21_{\text{sub.A}}$ of the organic EL element **21** is formed on the fourth wiring layer **84** that is the uppermost layer of the multilayer wiring structure.

Example of Forming Capacitor Element Using Wiring Layer Right Above Gate Electrode

(51) Here, in the multilayer wiring structure, as shown in FIG. 3, a case where the holding capacitor **24** is formed by using the first wiring layer **81** right above the gate electrode $22_{\text{sub.G}}$ of the driving transistor **22** will be considered as a conventional example. In the case of the conventional example, the first wiring layer **81** and the second wiring layer **82** form the holding capacitor **24** of a metal-insulator-metal (MIM) structure. In this case, in the first wiring layer **81** right above the gate electrode $22_{\text{sub.G}}$ of the driving transistor **22**, since the holding capacitor **24**

is formed avoiding other gate wiring or the like, it is difficult to efficiently form the holding capacitor **24**.

First Embodiment

(52) A first embodiment is an example in which the holding capacitor **24** having the MIM structure is formed using the anode electrode **21.sub.A** of the organic EL element **21**. FIG. **4** shows a sectional structure of the holding capacitor **24** according to the first embodiment.

(53) As shown in FIG. **4**, the holding capacitor **24** according to the first embodiment includes a first electrode **241** formed in a wiring layer (third wiring layer **83**) below the anode electrode **21.sub.A**, and a second electrode **243** formed opposite to the first electrode **241** and is electrically connected to the anode electrode **21.sub.A** via a contact unit **242**. Then, a dielectric **244** as a first insulating layer is interposed between the first electrode **241** and the second electrode **243**. The dielectric **244** includes a material (for example, SiO) for forming the fourth insulating layer **74** or a high dielectric material (for example, SiN) different from the fourth insulating layer **74**. As a result, the holding capacitor **24** having the MIM structure is formed using the anode electrode **21.sub.A** of the organic EL element **21**.

(54) As described above, the first electrode **241** is formed in a wiring layer below the anode electrode **21.sub.A**, and the second electrode **243** is electrically connected to the anode electrode **21.sub.A** via the contact unit **242**, so that a region below the anode electrode **21.sub.A** can be efficiently used to form the holding capacitor **24**. Furthermore, the length of the contact unit **242** is adjusted and an interval between the first electrode **241** and the second electrode **243** is arbitrarily set, so that the holding capacitor **24** having a desired capacitance value can be easily formed. As a result, deterioration in display quality due to insufficient capacity of the holding capacitor **24** can be suppressed.

Second Embodiment

(55) A second embodiment is a modification of the first embodiment, and is an example in which the holding capacitor **24** includes two capacitor elements. FIG. **5** shows a sectional structure of the holding capacitor **24** according to the second embodiment.

(56) In FIG. **5**, if the holding capacitor **24** according to the first embodiment is a first capacitor element **24.sub.1**, the holding capacitor **24** according to the second embodiment includes, in addition to the first capacitor element **24.sub.1**, a second capacitor element **24.sub.2**, and the first capacitor element **24.sub.1** and the second capacitor element **24.sub.2** are connected in parallel. The first capacitor element **24.sub.1** and the second capacitor element **24.sub.2** are connected in parallel, so that, in the case of the first capacitor element **24.sub.1** alone, in other words, the capacitance value of the holding capacitor **24** according to the second embodiment can be set to be larger than the holding capacitor **24** according to the first embodiment.

(57) Specifically, as similar to the holding capacitor **24** according to the first embodiment, the first capacitor element **24.sub.1** includes a first electrode **241** formed in a wiring layer (third wiring layer **83**) below the anode electrode **21.sub.A**, and a second electrode **243** formed opposite to the first electrode **241**. Then, the first capacitor element **24.sub.1** has the MIM structure in which a material for forming the fourth insulating layer **74** or the dielectric material **244** including a high-dielectric material different from the fourth insulating layer **74** is interposed between the first electrode **241** and the second electrode **243**. In the first capacitor element **24.sub.1** having the MIM structure, the length of the contact unit **242** is adjusted and an interval between the first electrode **241** and the second electrode **243** is arbitrarily set, so that the first capacitor element **24.sub.1** having a desired capacitance value can be easily formed.

(58) While the first capacitor element **24.sub.1** is formed by using the anode electrode **21.sub.A** of the organic EL element **21**, the second capacitor element **24.sub.2** is formed by using a wiring layer right below the anode electrode **21.sub.A** (the second wiring layer **82** and the third wiring layer **83**). Specifically, the second capacitor element **24.sub.2** includes a third electrode **245** formed in a wiring layer (second wiring layer **82**) below the first electrode **241**, and a fourth

electrode **247** formed opposite to the third electrode **245** and is electrically connected to the first electrode **241** via a contact unit **246**. Then, a dielectric **248** as a second insulating layer is interposed between the third electrode **245** and the fourth electrode **247**. The dielectric **248** includes a material (for example, SiO) for forming the third insulating layer **73** or a high dielectric material (for example, SiN) different from the third insulating layer **73**. In the second capacitor element **24.sub.2** having the MIM structure, the length of the contact unit **246** is adjusted and an interval between the third electrode **245** and the fourth electrode **247** is arbitrarily set, so that the second capacitor element **24.sub.2** having a desired capacitance value can be easily formed.

(59) In the second capacitor element **24.sub.2** having the above configuration, the third electrode **245** is electrically connected to the anode electrode **21.sub.A** via the contact unit **91**, the island shaped relay electrode **92**, and the contact unit **93**. Furthermore, the fourth electrode **247** is electrically connected to the second electrode **243** through a wiring (not shown). As a result, the second capacitor element **24.sub.2** is connected in parallel to the first capacitor **24.sub.1**.

(60) As described above, the holding capacitor **24** according to the second embodiment uses the anode electrode **21.sub.A** to form the first capacitor element **24.sub.1** having the MIM structure, and uses the wiring layer right below the anode electrode **21.sub.A** to form the second capacitor element **24.sub.2** having the MIM structure, and both elements **24.sub.1** and **24.sub.2** are connected in parallel. This makes it possible to efficiently use the region below the anode electrode **21.sub.A** to form the holding capacitor **24** and to set the capacitance value of the holding capacitor **24** according to the second embodiment to be larger than in the case of the first capacitor element **24.sub.1** alone. As a result, deterioration in display quality due to insufficient capacity of the holding capacitor **24** can be suppressed.

Third Embodiment

(61) A third embodiment is a modification of the second embodiment, and is an example in which only the second capacitor element **24.sub.2** of the second embodiment is used as the holding capacitor **24**. FIG. **6** shows a sectional structure of the holding capacitor **24** according to the third embodiment.

(62) As shown in FIG. **6**, the holding capacitor **24** according to third embodiment has a configuration including a capacitor element having the MIM structure formed using a wiring layer (the second wiring layer **82** and the third wiring layer **83**) right below the anode electrode **21.sub.A** of the organic EL element **21**.

(63) As described above, the holding capacitor **24** having the MIM structure using the wiring layer right below the anode electrode **21.sub.A** is formed, so that the region below the anode electrode **21.sub.A** can be efficiently used to form the holding capacitor **24** having a desired capacitance value as compared with the case of using the wiring layer (the first wiring layer **81**) right above the gate electrode **22.sub.G** to form the holding capacitor **24** (see FIG. **3**). As a result, deterioration in display quality due to insufficient capacity of the holding capacitor **24** can be suppressed as similar to the case of the first and second embodiments.

(64) <Modification>

(65) Although the technology of the present disclosure has been described above on the basis of the preferred embodiments, the technique of the present disclosure is not limited to the embodiments. The configuration and structure of the display device described in each of the above embodiments are illustrative and can be changed as appropriate. For example, in each of the above embodiments, a pixel circuit having a circuit configuration of 2Tr1C is exemplified. However, the pixel circuit is not limited to a circuit configuration of 2Tr1C, and a transistor (Tr) or a capacitor element (C) can be increased as needed.

(66) <Electronic Device of the Present Disclosure>

(67) The display device according to the present disclosure described above can be used as a display unit (display device) of an electronic device in any fields that displays a video signal input to an electronic device or a video signal generated in the electronic device as an image or video.

Examples of the electronic device can include a television set, a notebook personal computer, a digital still camera, a mobile terminal device such as a mobile phone, a head mount display, and the like. However, the electronic device is not limited to these.

(68) As described above, the following effects can be obtained by using the display device of the present disclosure as a display unit in electronic devices of any fields. In other words, according to the display device of the present disclosure, deterioration in display quality due to insufficient capacity of the holding capacitor **24** can be suppressed. Accordingly, the display device of the present disclosure is used as a display unit (display device) of the electronic device, and thereby, the display quality of the display image can be improved.

(69) The display device of the present disclosure also includes a module shape of a sealed configuration. Examples of such a display device include a display module formed by affixing a facing unit such as transparent glass to a pixel array unit. Note that the display module may be provided with a circuit unit for inputting and outputting a signal or the like from the outside to the pixel array unit, a flexible printed circuit (FPC), or the like. Hereinafter, a digital still camera and a head mounted display will be exemplified as specific examples of the electronic device using the display device of the present disclosure. However, the specific examples illustrated here are merely examples, and the present invention is not limited thereto.

Specific Example 1

(70) FIG. **7** is an external view of a lens interchangeable single lens reflex type digital still camera, FIG. **7A** is a front view thereof, and FIG. **7B** is a rear view thereof. A lens interchangeable single lens reflex type digital still camera has, for example, an interchangeable photographic lens unit (interchangeable lens) **112** on the front right side of a camera body unit (camera body) **111**, and a grip unit **113** that is gripped by a photographer on the front left side.

(71) Then, a monitor **114** is provided substantially at the center of the rear surface of the camera body unit **111**. An electronic view finder (eyepiece window) **115** is provided on the upper portion of the monitor **114**. By looking into an electronic view finder **115**, the photographer can visually recognize an optical image of a subject introduced from the photographing lens unit **112** and determine the composition.

(72) In the lens interchangeable single lens reflex type digital still camera having the above configuration, the display device of the present disclosure can be used as the electronic view finder **115**. In other words, the lens interchangeable single lens reflex type digital still camera according to the present example is manufactured by using the display device of the present disclosure as the electronic view finder **115**.

Specific Example 2

(73) FIG. **8** is an external view of a head mounted display. The head mount display has, for example, an ear hanger **212** worn in the head of a user, on both sides of a glasses-shaped display unit **211**. In this head mounted display, the display device of the present disclosure can be used as the display unit **211**. In other words, the head mounted display according to the present example is manufactured by using the display device of the present disclosure as the display unit **211**.

(74) <Configuration that the Present Disclosure can Have>

(75) Note that, the present disclosure can also have the following configuration. [1] A display device including a pixel arranged therein, the pixel including a light emitting unit in which an anode electrode is formed in the uppermost layer of a multilayer wiring structure formed by alternately stacking a plurality of insulating layers and a plurality of wiring layers and a capacitor element electrically connected to the anode electrode of the light emitting unit, the capacitor element including a first electrode formed in a wiring layer below the anode electrode, a second electrode formed opposite to the first electrode, and electrically connected to the anode electrode of the light emitting unit via a contact unit, and a first insulating layer interposed between the first electrode and the second electrode. [2] The display device described in [1] above, in which the capacitor element includes: a third electrode formed in a wiring layer below the first electrode; a

fourth electrode formed opposite to the third electrode, and electrically connected to the first electrode via the contact unit; and a second insulating layer interposed between the third electrode and the fourth electrode, the third electrode is electrically connected to the anode electrode, and the fourth electrode is electrically connected to the second electrode. [3] The display device described in [1] or [2] above, in which the pixel includes a driving transistor that drives the light emitting unit, and a writing transistor that writes a signal to a gate electrode of the driving transistor, and the capacitor element is connected between the gate electrode of the driving transistor and the anode electrode of the light emitting unit, and holds the signal written by the writing transistor. [4] The display device described in any of [1] to [3] above, in which a substrate on which the pixels are two-dimensionally arranged is a semiconductor substrate. [5] The display device described in any of [1] to [4] above, in which the light emitting unit includes an organic electroluminescence element. [6] An electronic device including a display device, the display device including a pixel arranged therein, the pixel including a light emitting unit in which an anode electrode is formed in the uppermost layer of a multilayer wiring structure formed by alternately stacking a plurality of insulating layers and a plurality of wiring layers, and a capacitor element electrically connected to the anode electrode of the light emitting unit, the capacitor element including a first electrode formed in a wiring layer below the anode electrode, a second electrode formed opposite to the first electrode, and electrically connected to the anode electrode of the light emitting unit via a contact unit, and a first insulating layer interposed between the first electrode and the second electrode.

REFERENCE SIGNS LIST

(76) **10** Organic EL display device **20** Pixel (pixel circuit) **21** (**21R**, **21G**, **21B**) Organic EL element **22** Driving transistor **23** Writing transistor **24** Holding capacitor **30** Pixel array unit **31** (**31.sub.1** to **31.sub.m**) Scanning line **32** (**32.sub.1** to **32.sub.m**) Power supply line **33** (**33.sub.1** to **33.sub.n**) Signal line **34** Common power supply line **40** Writing scanning unit **50** Power supply scanning unit **60** Signal output unit **70** Display panel **81** First wiring layer **82** Second wiring layer **83** Third wiring layer **84** Fourth wiring layer

Claims

1. A display device comprising: a plurality of pixel circuits arranged in a matrix, wherein each of the pixel circuits includes: a light emitting element; a capacitance component; a sampling transistor configured to supply a data voltage supplied from a data signal line to the capacitance component; and a drive transistor configured to supply driving current to the light emitting element according to a voltage stored in the capacitance component, wherein the capacitance component includes a first capacitive element and a second capacitive element connected in parallel to the first capacitive element, an anode electrode is formed in a first layer, the first capacitive element includes a first electrode formed in a second layer arranged under the first layer and connected to a gate of the drive transistor, and a second electrode opposed to the first electrode, the second capacitive element includes a third electrode formed in a third layer arranged under the second layer, and a fourth electrode opposed to the third electrode and connected to the first electrode, and the second electrode and the third electrode are electrically directly connected to the anode electrode.
2. The display device according to claim 1, wherein the second electrode is electrically connected to the third electrode.
3. The display device according to claim 1, wherein the first electrode is electrically connected to the fourth electrode via a plurality of contacts.
4. The display device according to claim 1, wherein the fourth electrode is formed below the first electrode.
5. The display device according to claim 1, wherein at least a part of a first region where a gate of the drive transistor is formed overlaps with at least a part of a second region where the first electrode is formed in a plan view.

6. The display device according to claim 1, wherein at least a part of a first region where a gate of the drive transistor is formed overlaps with at least a part of a third region where the second electrode is formed in a plan view.
 7. The display device according to claim 1, wherein at least a part of a second region where the first electrode is formed overlaps with at least a part of a fourth region where the third electrode is formed in a plan view.
 8. The display device according to claim 1, wherein at least a part of a second region where the first electrode is formed overlaps with at least a part of a fifth region where the fourth electrode is formed in a plan view.
 9. The display device according to claim 1, wherein each of the first capacitive element and the second capacitive element is a MIM (Metal-Insulator-Metal) capacitor.
 10. An electronic apparatus comprising the display device according to claim 1.
 11. The electronic apparatus according to claim 10, wherein the second electrode is electrically connected to the third electrode.
 12. The electronic apparatus according to claim 11, wherein the second electrode and the third electrode are electrically connected to the anode electrode.
 13. The electronic apparatus according to claim 10, wherein the first electrode is electrically connected to the fourth electrode via a plurality of contacts.
 14. The electronic apparatus according to claim 10, wherein the fourth electrode is formed below the first electrode.
 15. The electronic apparatus according to claim 10, wherein at least a part of a first region where a gate of the drive transistor is formed overlaps with at least a part of a second region where the first electrode is formed in a plan view.
 16. The electronic apparatus according to claim 10, wherein at least a part of a first region where a gate of the drive transistor is formed overlaps with at least a part of a third region where the second electrode is formed in a plan view.
 17. The electronic apparatus according to claim 10, wherein at least a part of a second region where the first electrode is formed overlaps with at least a part of a fourth region where the third electrode is formed in a plan view.
 18. The electronic apparatus according to claim 10, wherein at least a part of a second region where the first electrode is formed overlaps with at least a part of a fifth region where the fourth electrode is formed in a plan view.
 19. The electronic apparatus according to claim 10, wherein each of the first capacitive element and the second capacitive element is a MIM (Metal-Insulator-Metal) capacitor.
-